CHANGKEER

Self-Service Mobile Ordering, Dynamic Locked-Amount QRIS Payments, Real-Time Kitchen Display System (KDS), and Store Management.

PROJECT TEAM

LEAD ARCHITECT

ORGANIZATION / BRAND

ACADEMIC YEAR / RELEASE

Operational Friction in the Artisan Tea Sector

Why traditional cafe counters fail to meet modern guest expectations and barista precision.

Core Frictions in Traditional Tea Stores

- **Exhausting Counter Queues:** Guests stand in long lines simply to decide on custom botanical blends, degrading the relaxing tea lounge experience.
- **Costly Typos on Static QR Codes:** Static printed paper stickers require guests to manually key in prices, triggering frequent underpayments or double transfers.
- **Damaged Paper Tickets at the Brew Bar:** Paper kitchen chits easily get soaked with hot water or lost, leading to swapped orders and ruined steep times.
- **Lack of Recipe Calibration:** Herbal extraction varies wildly between baristas without automated digital recipe guidance.

EVERYDAY ANALOGY: CHAOS BEHIND THE BAR

Imagine a barista carefully maintaining water temperature at 85°C for mountain jasmine. Suddenly, the cashier yells a custom 30% sweetness modification, while the paper receipt slips into a puddle of spilled tea. The barista loses track of the timer, and the brew turns bitter and astringent. This is the exact issue CHANGKEER eliminates.

THE STRATEGIC SOLUTION

Eliminating 100% of paper tickets at the bar, preventing financial transfer errors through dynamic QRIS, and empowering guests to configure their cups right from their seats.

The CHANGKEER Solution: 24 Seasons Botanical Teas

Harmonizing bodily metabolism with Indonesia's rich botanical heritage and cosmic natural cycles.

CHANGKEER presents an authentic herbal tea sanctuary celebrating the ancient 24 Solar Terms calendar. Each cup is intentionally designed to align bodily vitality with prevailing weather and atmospheric conditions.

LAYMAN EXPLANATION: WHAT IS AN HERBAL TISANE?

A tisane is a pure botanical infusion created from whole herbs, spices, and whole flower blossoms (such as dried rose, mountain jasmine, blue pea flower, and ginger root). Unlike mass-market tea bags that often use synthetic aromatics, CHANGKEER Tisanes are 100% natural, preservative-free, and stomach-friendly.

SEASON	SENSORY PROFILE	BODILY WELLNESS FUNCTION
Spring	Fresh Floral Bouquet	Refreshes mental vitality & aids gentle seasonal detox.
Summer	Cooling Mint & Citrus	Lowers internal heat & delivers intense hydration.
Autumn	Chamomile & Cinnamon	Soothes nervous tension & relieves daily stress.
Winter	Aged Red Ginger & Clove	Provides deep thermal comfort & strengthens immunity.

Three Signature Botanical Blends

Whole-leaf loose teas and premium Indonesian botanicals steeped with scientific precision.



Jasmine Tea Signature

High-altitude green tea scented with natural jasmine petals. Calms mental fatigue and provides a silky, delicate finish.



Vanilla Mint Tea

Crisp highland peppermint balanced with sweet organic vanilla extract. Clears breathing passages and cools the body.



English Breakfast Tea

Robust, full-bodied Indonesian black tea providing sustained morning focus. 100% bleach-free, microplastic-free pouches.

Frictionless Mobile Ordering & Customization

Empowering guests to tailor their tea ritual without downloading any third-party app.

The Web-Based Guest Journey

- **No App Installation Required:** Guests scan a stylish table QR code using their native phone camera to instantly access the live catalog.
- **Temperature Selection:** Traditional Hot Clay Pot (warm comfort) or Refreshing Iced Shaken Tea (+IDR 2,000 with craft ice).
- **5 Calibrated Sweetness Levels:** 0% (Unsweetened), 30% (Light), 50% (Tea Forward), 75% (Balanced), and 100% (Full Sweetness).
- **Botanical Petal Add-Ons (+IDR 3,000):** Fresh Butterfly Pea, Dried Mountain Jasmine, Royal Marigold, or Raw Forest Honey.

WHY GUESTS LOVE COMPLETE PERSONALIZATION

Every guest has distinct wellness needs: diabetics require 0% sugar brews, fatigued individuals need raw forest honey, and lifestyle seekers love iced teas infused with purple butterfly pea petals. The CHANGKEER system accommodates every preference in seconds.

End-to-End Automated Transaction Pipeline

Complete digital automation from menu exploration to the barista's pickup counter.

1
Configure on Phone
Guest picks tea blend, temperature, sweetness & toppings.

→

2
Dynamic QRIS
Server locks in the exact payable bill amount into barcode.

→

3
Security Check
Backend re-validates catalog prices & payment state.

→

4
Barista KDS Ticket
Touchscreen display updates with order queue in real time.

→

5
Precision Handover
Barista steeps tea accurately & delivers cup to table.

LAYMAN EXPLANATION: THE SUPERIORITY OF DYNAMIC QRIS

On traditional paper QR codes, guests must manually type in "IDR 28,000". If they accidentally type "IDR 2,800", the store loses money. With CHANGKEER Dynamic QRIS, the scanned barcode already carries the exact locked IDR 28,000 amount. Guests only tap 'Confirm' on their mobile banking app.

Financial Protection & Anti-Tampering Shield

Multi-tier safeguards protecting both retail revenues and consumer trust.

1. Server-Side Anti-Tampering

All product unit prices, ice surcharges (+IDR 2,000), and flower add-ons are strictly re-verified on the central backend, never trusted to browser variables.

ANALOGY: THE INCORRUPTIBLE CASHIER

If a dishonest customer tampers with browser code to change an IDR 35,000 premium blend to IDR 1,000, the central server instantly flags the discrepancy and aborts the transaction immediately.

2. Idempotency (Double-Charge Prevention)

When network connectivity stutters and a customer taps "Pay" multiple times, the transaction lock ensures only the first request is executed.

ANALOGY: THE FAIR TICKET MACHINE

Guarantees that the customer's bank account or e-wallet balance is never debited twice, no matter how many times the button was pressed.

Kitchen Display System (KDS): A Paperless Bar

Transforming the tea preparation counter into an organized, hygienic brewing laboratory.

Operational Value for Baristas

- **Chronological Order Queue:** Orders placed earlier sit automatically at the front of the queue, preventing fulfillment delays.
- **Vibrant Color-Coded States:** Amber (Waiting), Blue (Active Extraction), and Green (Ready for Pickup).
- **Crystal-Clear Customization:** Sweetness doses (ml) and flower toppings are clearly highlighted on screen.
- **Zero Paper Clutter:** Brewing surfaces remain completely dry, sterile, and professional.

TEA CATEGORY	WATER TEMPERATURE	STANDARD STEEPING DURATION
Green Tea & Jasmine	80°C – 85°C	3.5 – 4 Minutes (Delicate)
Robust Black Tea	95°C – 98°C	3.5 Minutes (Bold)
Whole Floral Tisanes	90°C – 95°C	5 Minutes (Full Color)
Root & Ginger Tisanes	98°C – 100°C	6 Minutes (Active Oils)

Next-Generation Engineering Stack

Engineered with high-speed modern web infrastructure to ensure uninterrupted retail uptime.

Next.js 16 & React 19

Ultra-fast catalog rendering on mobile devices with negligible cellular data usage.

Supabase Cloud & RLS

Encrypted PostgreSQL database with strict row-level authorization on customer orders.

Dual-Tier Resilience

If internet connection drops, the local memory store ensures counter POS continues smoothly.

MEASURABLE RETAIL IMPACT

Reduces customer ordering wait times by up to 65%, cuts tea beverage preparation errors to near 0%, and permanently eliminates paper receipt procurement overhead.

Engineering Team & Open Discussion

Empowering the future of Indonesian botanical tea rituals through disciplined digital engineering.

Project Development Team Roster

DEPARTMENT / ROLE	LEAD PERSON
Project Lead & System Architect	**Muhammad Ilham**
Frontend & UI/UX Division	User Interface Team
Botanical R&D & Barista Standards	Botanical Research Team
Store Operations & Cashier Services	Store Operations Team

FINAL SYNTHESIS

CHANGKEER proves that heritage herbal craftsmanship can seamlessly harmonize with cutting-edge web technology. The platform is ready for commercial multi-store deployment.

Thank You for Your Consideration

We warmly invite questions, critique, and discussion.

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